Abu Dhabi City Campus Daily/Weekly Planner

Week 8: 19th – 22nd October 2026

Subject: Mathematics Pre-AP Geometry with Statistics **Unit Title: Measurement in Data** **Grade: 9**

Lessons	Learning targets	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1: **Conditional Probability**	I can calculate conditional probability by restricting the sample space to the condition that has already occurred.	1.2 Provide concrete outcome cards/manipulatives before symbolic notation. 2.3 Represent probability using lists, tables, fractions, decimals, and probability scales.	Pre-AP Geometry with Statistics Course Guide, Key Concept 1.3.1. Pre-AP Geometry, Lesson 1.2.3 Calculate relative frequencies, joint frequencies, marginal frequenci Pre-AP Geometry, Lesson 1.2.4 Determine if two events are independent. Pre-AP Geometry, Lesson 1.3.1 Distinguish between accuracy and precision as measures of statistical Pre-AP Geometry, Lesson 1.3.4 Identify biases in sampling methods for experiments, observational st
Lesson 2: **Independent Events**	I can determine whether two events are independent by comparing how one event affects the probability of the other.	8.2 Provide accessible item sequencing while maintaining grade-level reasoning demand.	
Lesson 3: **Accuracy, Precision & Variability**	I can distinguish accuracy from precision and use variability to describe how consistent a set of measurements is.	5.3 Use graduated examples before independent classification and interpretation. 2.3 Represent measurement quality using target diagrams, numerical summaries and contextual statements.	
Lesson 4: **Statistical Bias**	I can identify systematic bias and explain how it can cause measurements to consistently overestimate or underestimate a true value.	5.3 Use worked examples before independent scenario analysis. 2.3 Represent bias through diagrams, numerical examples and contextual explanations.	
Homework			
Teacher's notes / Reminders	Students should bring a device, scientific calculator and notebook and have access to the PPTs.		

Technology Integration	Microsoft Teams and OneNote; GDC/spreadsheet; two-way tables; probability trees; measurement data sets; target-board visuals; digital simulations.

Cross-curricular links	**Science:** repeated measurements, experimental error, reliability and calibration. **Business/Economics:** evidence-based decisions and the effect of biased data on conclusions.
External Assessment Integration	Direct alignment with College Board Pre-AP
Wellbeing/ ATL Skills	**Thinking skills:** calculate, compare, analyse relationships, distinguish accuracy from precision, identify systematic error and evaluate evidence. **Communication skills:** use precise probability/statistical vocabulary and justify conclusions with tables, calculations and context. **Self-management skills:** organise multi-step calculations, check reasonableness, manage investigation data and act on feedback. **Social skills:** compare strategies, discuss interpretations and provide evidence-based peer feedback.
Standards	Pre-AP Geometry with Statistics – Unit 1: Measurement in Data Key Concept 1.2: Chance Events — 1.2.3 Calculate relative, joint, marginal and conditional frequencies/probabilities; 1.2.4 Determine whether two events are independent. Key Concept 1.3: Inferences from Data — 1.3.1 Distinguish accuracy and precision as measures of variability and distinguish statistical bias in measurements.
Inquiry/ TOK Question	How does the information we know change the probability we assign to an event, and how does the quality of a measurement process affect the confidence we can place in a statistical claim?
Success Criteria (Measurable)	**Lesson 1: Conditional Probability** • **All students must** identify the restricted sample space and calculate a conditional probability from a two-way table. • **Most students should** explain why the denominator changes when a condition is given and interpret the conditional probability in context. • **Some students could** compare P(A|B) and P(B|A), explain why they may be different, and justify their reasoning using a two-way table or probability model. **Lesson 2: Independence** • **All students must** determine whether two events are independent by comparing P(A|B) with P(A). • **Most students should** justify independence/dependence using probability evidence and explain the conclusion in context. • **Some students could** verify independence using a second method, such as $P(A \cap B)=P(A)P(B)$, and explain the real-world meaning. **Lesson 3: Accuracy, Precision & Variability** • **All students must** distinguish accuracy from precision. • **Most students should** use centre and variability evidence to explain a measurement set. • **Some students could** analyse a set of measurements and justify whether it is accurate, precise, both or neither.

	Lesson 4: Statistical Bias • **All students must** identify a biased measurement process. • **Most students should** explain the direction and effect of the bias. • **Some students could** propose a change to the measurement process that would reduce the bias while preserving the purpose of the investigation.
National Identity Domain	Belonging / Citizenship: Use UAE-relevant school, science and community measurement contexts so students see statistics as a tool for responsible decision-making.
	Area: Responsible decision making.
	Implementation: Students evaluate whether evidence is sufficiently accurate, precise and unbiased before accepting a statistical claim or making a decision.